

Linux Process Management Commands

This document explains useful Linux commands for monitoring and managing processes, with examples and outputs.



1. View All Processes

Command:

```
ps aux
```

- **a** → show processes for all users
- **u** → show user/owner of process
- **x** → show processes not attached to a terminal

Example Output:

```
angel@angel-VirtualBox:~$ pwd
/home/angel
angel@angel-VirtualBox:~$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.2	0.2	23316	14000	?	Ss	09:16	0:09	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	09:16	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	09:16	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/R-rcu_gp]
root	5	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/R-sync_wq]
root	6	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/R-kvfree_rcu_reclaim]
root	7	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/R-slub_flushwq]
root	8	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/R-netns]
root	11	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/0:0H-kblockd]
root	13	0.0	0.0	0	0	?	I<	09:16	0:00	[kworker/R-mm_percpu_wq]
root	14	0.0	0.0	0	0	?	I	09:16	0:00	[rcu_tasks_kthread]
root	15	0.0	0.0	0	0	?	I	09:16	0:00	[rcu_tasks_rude_kthread]
root	16	0.0	0.0	0	0	?	I	09:16	0:00	[rcu_tasks_trace_kthread]
root	17	0.0	0.0	0	0	?	S	09:16	0:02	[ksoftirqd/0]
root	18	0.6	0.0	0	0	?	I	09:16	0:24	[rcu_preempt]
root	19	0.0	0.0	0	0	?	S	09:16	0:00	[rcu_exp_par_gp_kthread_worker/0]
root	20	0.0	0.0	0	0	?	S	09:16	0:00	[rcu_exp_gp_kthread_worker]
root	21	0.0	0.0	0	0	?	S	09:16	0:00	[migration/0]
root	22	0.0	0.0	0	0	?	S	09:16	0:00	[idle_inject/0]
root	23	0.0	0.0	0	0	?	S	09:16	0:00	[cpuhp/0]
root	24	0.0	0.0	0	0	?	S	09:16	0:00	[cpuhp/1]
root	25	0.0	0.0	0	0	?	S	09:16	0:00	[idle_inject/1]
root	26	0.0	0.0	0	0	?	S	09:16	0:00	[migration/1]
root	27	0.0	0.0	0	0	?	S	09:16	0:02	[ksoftirqd/1]
root	30	0.0	0.0	0	0	?	S	09:16	0:00	[cpuhp/2]
root	31	0.0	0.0	0	0	?	S	09:16	0:00	[idle_inject/2]
root	32	0.0	0.0	0	0	?	S	09:16	0:00	[migration/2]
root	33	0.1	0.0	0	0	?	S	09:16	0:05	[ksoftirqd/2]
root	36	0.0	0.0	0	0	?	S	09:16	0:00	[cpuhp/3]



2. Process Tree

Command:

```
pstree -p
```

Shows parent-child relationships.

Example Output:

```
systemd(1)─NetworkManager(778)
           └─sshd(895)─sshd(1023)─bash(1024)─pstree(1101)
           └─mysqld(2001)
           └─python3(1234)
```

```
angel@angel-VirtualBox:~$ pstree -p
systemd(1)─ModemManager(1043)─{ModemManager}(1051)
           │                 │{ModemManager}(1055)
           │                 │{ModemManager}(1060)
           └─NetworkManager(1351)─{NetworkManager}(1352)
           │                     │{NetworkManager}(1353)
           │                     │{NetworkManager}(1354)
           └─accounts-daemon(951)─{accounts-daemon}(1005)
           │                   │{accounts-daemon}(1006)
           │                   │{accounts-daemon}(1008)
           └─avahi-daemon(925)─avahi-daemon(963)
           └─colord(1750)─{colord}(1755)
           │             │{colord}(1756)
           │             │{colord}(1758)
           └─cron(952)
           └─cups-browsed(1477)─{cups-browsed}(1489)
           │                 │{cups-browsed}(1490)
           │                 │{cups-browsed}(1491)
           └─cupsd(1174)
           └─dbus-daemon(926)
           └─fwupd(11241)─{fwupd}(11292)
           │             │{fwupd}(11384)
           │             │{fwupd}(11385)
           │             │{fwupd}(11386)
           │             │{fwupd}(11388)
           └─gdm3(1599)─gdm-session-wor(2187)─gdm-wayland-ses(2445)─gnome-session-b(2458)─{gnome-session-b}(252+
           │                                     │{gnome-session-b}(253+
           │                                     │{gnome-session-b}(253+
           │                                     │{gdm-wayland-ses}(2449)
           │                                     │{gdm-wayland-ses}(2450)
           │                                     │{gdm-wayland-ses}(2455)
```

3. Real-Time Monitoring

Command:

```
top
```

Displays live system resource usage.

```
angel@angel-VirtualBox:~$ top

top - 10:28:06 up 1:11, 1 user, load average: 0.22, 0.29, 0.58
Tasks: 230 total, 2 running, 228 sleeping, 0 stopped, 0 zombie
%Cpu(s): 14.4 us, 17.3 sy, 0.1 ni, 65.5 id, 0.1 wa, 0.0 hi, 2.5 si, 0.0 st
MiB Mem : 4943.9 total, 290.4 free, 2093.7 used, 2588.1 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 2850.2 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 2652 angel      20   0 5189908 486908 159300 S 42.4   9.6   27:14.19 gnome-shell
 13171 angel      20   0 3826388 234892 115976 R 37.7   4.6   0:01.14 gnome-character
13167 angel      20   0 1012392 51788 40396 S 18.9   1.0   0:00.57 nautilus
13173 angel      20   0 566008 49024 32336 S 16.2   1.0   0:00.49 gnome-clocks
 2380 angel      20   0 11484 6980 4452 S 8.6    0.1   0:05.77 dbus-daemon
13177 angel      20   0 498992 30220 23308 S 7.9    0.6   0:00.24 gnome-control-c
13169 angel      20   0 527024 17168 15248 S 3.0    0.3   0:00.09 gnome-calculato
 18 root        20   0 0 0 0 I 2.0    0.0   0:25.28 rcu_preempt
 3500 angel      20   0 701604 14272 11584 S 1.7    0.3   0:01.32 xdg-desktop-por
 2853 angel      20   0 388848 12152 6844 S 1.3    0.2   0:29.95 ibus-daemon
12847 root        20   0 0 0 0 I 1.3    0.0   0:01.09 kworker/u16:3-events_unbound
 267 root        19  -1 50848 17616 16336 S 1.0    0.3   0:02.22 systemd-journal
12803 angel      20   0 2876048 61880 46992 S 1.0    1.2   0:01.13 gjs
 1 root        20   0 23316 14000 9264 S 0.7    0.3   0:09.91 systemd
 988 syslog      20   0 222564 5844 4436 S 0.7    0.1   0:00.64 rsyslogd
2382 angel      20   0 316936 9952 8928 S 0.7    0.2   0:00.46 gnome-keyring-d
2556 angel      20   0 314268 7768 7128 S 0.7    0.2   0:00.34 gvfsd
3109 angel      20   0 389376 7796 6900 S 0.7    0.2   0:01.03 gvfs-afc-volume
3179 angel      20   0 236768 7292 6780 S 0.7    0.1   0:09.69 ibus-engine-sim
3526 angel      20   0 909136 41080 31096 S 0.7    0.8   0:00.99 xdg-desktop-por
10005 angel      20   0 11.4g 636020 239336 S 0.7   12.6   4:39.72 firefox
11134 angel      20   0 626024 56872 45412 S 0.7    1.1   0:06.17 gnome-terminal-
```

Example Output:

```
top - 10:20:51 up 2 days, 3:12, 2 users, load average: 0.22, 0.33, 0.45
Tasks: 197 total, 1 running, 196 sleeping, 0 stopped, 0 zombie
%Cpu(s): 12.3 us, 5.4 sy, 0.0 ni, 80.1 id, 2.2 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 8045632 total, 3564980 free, 1876324 used, 2604328 buff/cache

  PID  USER    PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 1234  vibhu    20   0 274532 15632 7892 R 45.0  1.5   0:12.34 python3
2001  mysql    20   0 450000 20988 7564 S 25.0  2.0   1:02.11 mysqld
```

👉 Press **q** to quit.

⚡ 4. Adjust Process Priority

Start process with low priority:

```
nice -n 10 sleep 300 &
```

Output:

```
[1] 13749
```

👉 Process with PID **13749** runs with nice value 10.

Change priority:

```
renice -n -5 -p 13749
```

Output:

```
13749 (process ID) old priority 10, new priority -5
```

```
angel@angel-VirtualBox:~$ nice -n 10 sleep 300 &  
[1] 13749  
angel@angel-VirtualBox:~$ sudo renice -n -5 -p 13749  
13749 (process ID) old priority 10, new priority -5  
angel@angel-VirtualBox:~$
```

5. CPU Affinity

Show which CPUs process can use:

```
taskset -cp 3050
```

Output:

```
pid 3050's current affinity list: 0-3
```

Restrict to core 1:

```
taskset -cp 1 3050
```

Output:

```
pid 3050's current affinity list: 1
```

6. I/O Scheduling Priority

```
ionice -c 3 -p 3050
```

Output:

```
successfully set pid 3050's IO scheduling class to idle
```

👉 Class 3 = only gets I/O when system is idle.

7. File Descriptors Used by a Process

Command: `lsof -p 3050 | head -5`

Example Output:

COMMAND	PID	USER	FD	TYPE	DEVICE	SIZE/OFF	NODE	NAME
sleep	3050	angel	cwd	DIR	253,0	4096	131073	/home/angel
sleep	3050	angel	rtd	DIR	253,0	4096	2	/
sleep	3050	angel	txt	REG	253,0	17520	133580	/usr/bin/sleep

8. Trace System Calls of a Process

Command: `strace -p 3050`

Example Output:

```
strace: Process 3050 attached
restart_syscall(<... resuming interrupted nanosleep ...>) = 0
nanosleep({tv_sec=300, tv_nsec=0}, 0x7ffd4a60d8b0) = ?
ERESTART_RESTARTBLOCK (Interrupted by signal)
```

👉 Great for debugging system calls and process behavior.

9. Find Process Using a Port

Command: `sudo fuser -n tcp 8080`

Output:

```
8080/tcp:          4321
```

👉 Process PID 4321 is using port 8080.

10. Per-Process Statistics

Command: `pidstat -p 3050 2 3`

Example Output:

Linux 5.15.0 (ubuntu)	09/25/25	_x86_64_		(4 CPU)		
12:30:20	UID	PID	%usr	%system	%CPU	CPU Command
12:30:22	1000	3050	0.00	0.00	0.00	1 sleep
12:30:24	1000	3050	0.00	0.00	0.00	1 sleep
12:30:26	1000	3050	0.00	0.00	0.00	1 sleep

🔊 Displays CPU usage every 2 seconds, 3 times.



11. Control Groups (cgroups) for Resource Limits

```
Create a New cgroup:
sudo cgcreate -g cpu,memory:/testgroup
Limit CPU and Memory:
echo 50000 | sudo tee /sys/fs/cgroup/cpu/testgroup/cpu.cfs_quota_us
echo 100M | sudo tee
/sys/fs/cgroup/memory/testgroup/memory.limit_in_bytes
Add Process (PID 3050) to cgroup:
echo 3050 | sudo tee /sys/fs/cgroup/cpu/testgroup/cgroup.procs
```



12. Alternatives to nice / renice

```
1. chrt (Real-Time Scheduling)
Set real-time scheduling policies (FIFO or Round Robin).
sudo chrt -f 50 sleep 1000
chrt -p <pid>

2. ionice (I/O Priority Control)
ionice -c 2 -n 7 tar -czf backup.tar.gz /home

3. taskset (CPU Affinity)
taskset -c 1 firefox

4. Control Groups (cgroups)
sudo cgcreate -g cpu,memory:/lowprio
echo 20000 | sudo tee /sys/fs/cgroup/cpu/lowprio/cpu.cfs_quota_us
echo 200M | sudo tee /sys/fs/cgroup/memory/lowprio/memory.limit_in_bytes
echo 1234 | sudo tee /sys/fs/cgroup/cpu/lowprio/cgroup.procs

5. systemd-run
systemd-run --scope -p CPUWeight=200 stress --cpu 4

6. schedtool
sudo schedtool -R -p 10 <pid>
```



Summary Table

Tool Focus Alternative to chrt Real-time scheduling policies nice ionice I/O priority control (complementary) taskset CPU affinity control (complementary) cgroups Fine-grained resource management nice (more powerful) systemd-run systemd + cgroups resource mgmt nice schedtool Custom scheduling policies nice